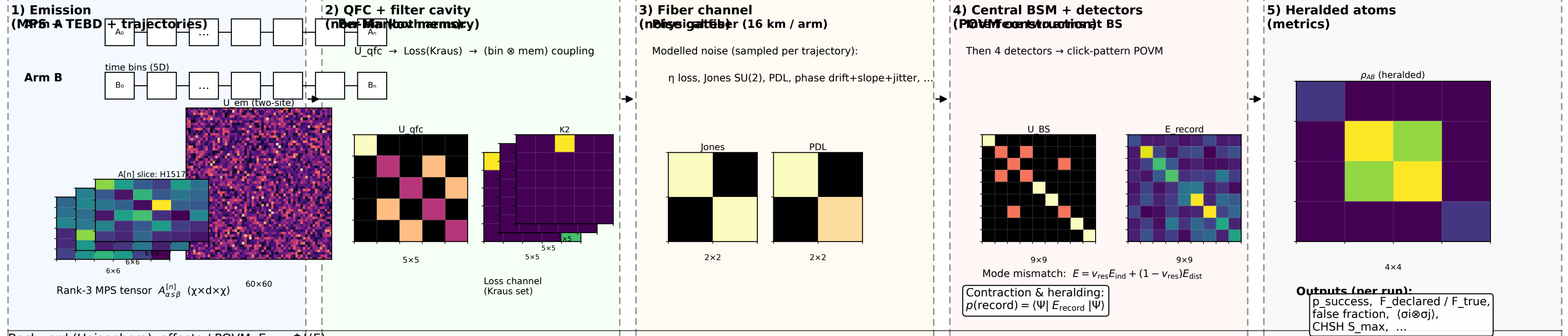
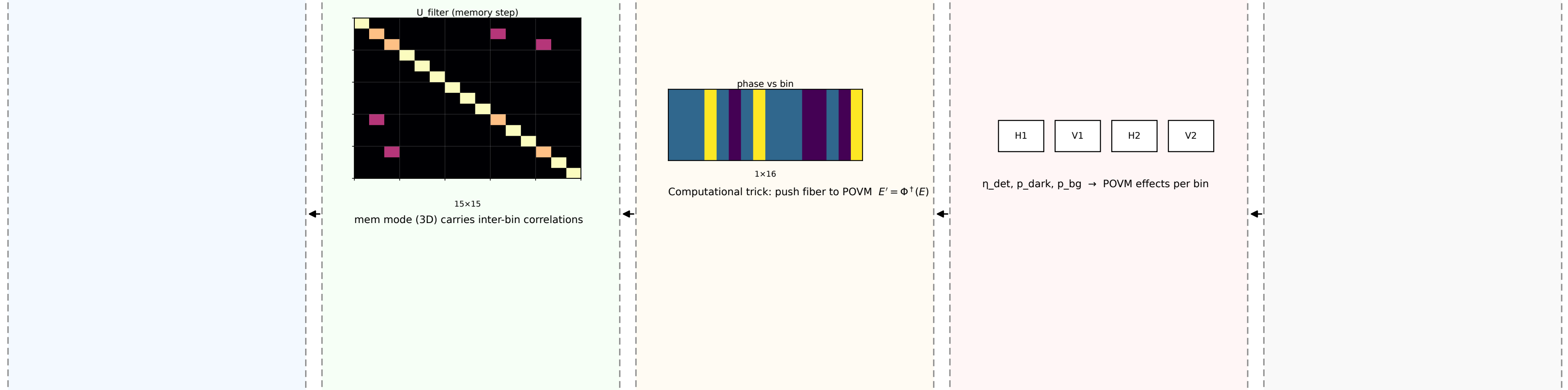


End-to-end time-bin MPS simulation: Rb cavity emission $\rightarrow$ QFC/filter $\rightarrow$ fiber noise (pushed to POVM) $\rightarrow$ BSM $\rightarrow$ heralded atoms

Forward (Schrödinger): state / MPS evolution



Backward (Heisenberg): effects / POVM $E \leftarrow \Phi^\dagger(E)$



Heatmaps show |matrix elements|; stacked heatmaps indicate Kraus sets / tensor slices.